

Write your full name at the top of each page. Write your answers in the space provided. Use the back of the page if needed, but be sure to use arrows to indicate that you have done so. **Please show your work!**

1. **TFUs.** Determine whether the following statements are **true**, **false**, or **uncertain**, and justify your answer in no more than one paragraph. Be concise. [10 points each]

- (a) In a steady state of the Ramsey-Cass-Koopmans model, the real interest rate r^{ss} must be strictly positive. (*Hint:* Think about the steady-state Euler equation.)

True. The steady-state Euler equation implies $\beta(1 + r^{\text{ss}}) = 1$. Since $\beta < 1$ (else the household's lifetime objective would not be well-defined), it must be that $r^{\text{ss}} > 0$.

- (b) The following production function exhibits constant returns to scale in the tuple of inputs $\{K, L, h\}$:

$$Y = K^\alpha (AhL)^{1-\alpha}.$$

False. For any $\lambda > 0$,

$$\begin{aligned} F(\lambda K, \lambda L, \lambda h) &= (\lambda K)^\alpha [A(\lambda h)(\lambda L)]^{1-\alpha} \\ &= \lambda^{2-\alpha} K^\alpha [AhL]^{1-\alpha} \\ &= \lambda^{2-\alpha} F(K, L, h) \\ &\neq \lambda F(K, L, h). \end{aligned}$$

If we instead thought about the tuple $\{K, h \times L\}$, treating quality-adjusted labor as one input, then the production function would exhibit constant returns to scale. This is the same distinction we made with labor-augmenting productivity A in the notes.

- (c) In a competitive model where a representative firm uses the production function in (b), the wage rate can be written

$$\ln w = \Lambda + \ln h$$

for some Λ that is a function of parameters and other variables. (*Hint:* If it were true, the ratio K/Y would be useful.)

True. From the firm's first-order condition, $w = (1 - \alpha) \frac{Y}{L}$. We can write GDP per worker as

$$\begin{aligned} \frac{Y}{L} &= \frac{K^\alpha (AhL)^{1-\alpha}}{L} = \left(\frac{K}{AhL} \right)^\alpha Ah = \left(\frac{K^{1-\alpha}}{(AhL)^{1-\alpha}} \right)^{\frac{\alpha}{1-\alpha}} Ah \\ &= \left(\frac{K}{K^\alpha (AhL)^{1-\alpha}} \right)^{\frac{\alpha}{1-\alpha}} Ah \\ &= \left(\frac{K}{Y} \right)^{\frac{\alpha}{1-\alpha}} Ah. \end{aligned}$$

Taking logs,

$$\begin{aligned} \ln w &= \ln(1 - \alpha) + \ln(Y/L) \\ &= \ln(1 - \alpha) + \frac{\alpha}{1 - \alpha} \ln(K/Y) + \ln(A) + \ln h \\ &\equiv \Lambda + \ln h. \end{aligned}$$

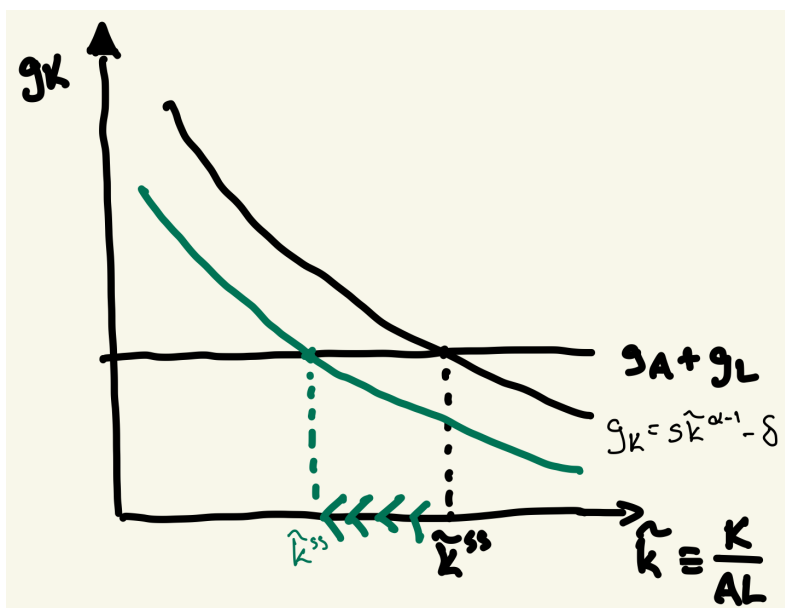
2. **Comparative statics.** For each of the following scenarios, assume that the economy begins in a steady state of the Solow model:

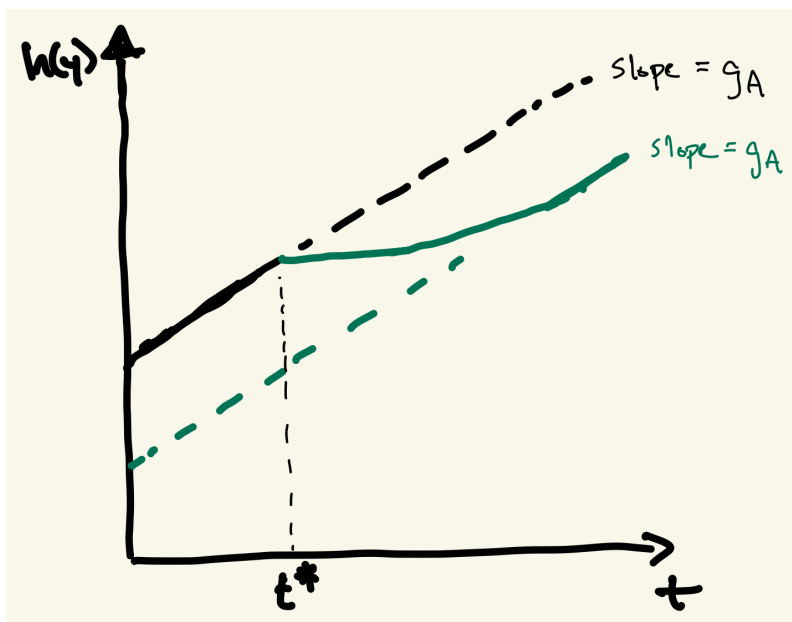
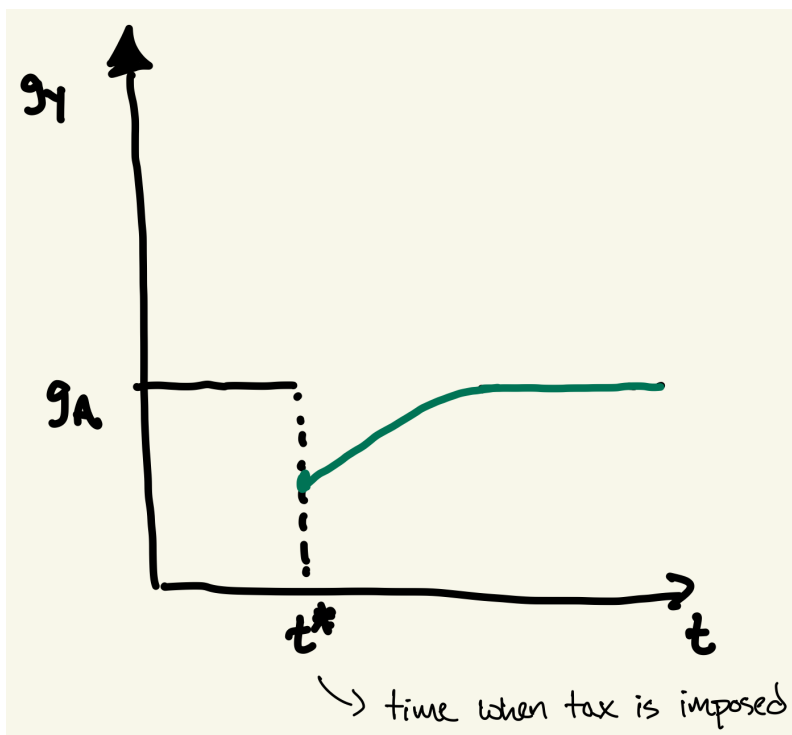
$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}, \quad \dot{K}_t = I_t - \delta K_t, \quad g_A, g_L > 0.$$

For each scenario, draw three figures showing how each of the following terms evolve over time: the amount of capital per efficiency unit of labor ($\tilde{k}_t$), the growth rate of GDP per capita (g_y), and the log of GDP per capita ($\ln(y_t)$). Be sure to clearly label all axes and curves in your figures.

- (a) The government decides to levy an income tax on both wage and capital income at the rate τ . This means that the amount spent on gross capital formation is now $I_t = s(1 - \tau)Y_t$ rather than $I_t = sY_t$. Suppose first that tax revenues are simply consumed by the government. [15 points]

Answer. The effective saving rate is now $\bar{s} = s(1 - \tau) < s$. This is the same problem as Problem Set 2, Question 2(a).





- (b) Now suppose instead that the taxes were spent by the government on purchasing new capital goods (e.g., infrastructure), so that capital accumulation is

$$\dot{K}_t = I_t + \tau Y_t - \delta K_t.$$

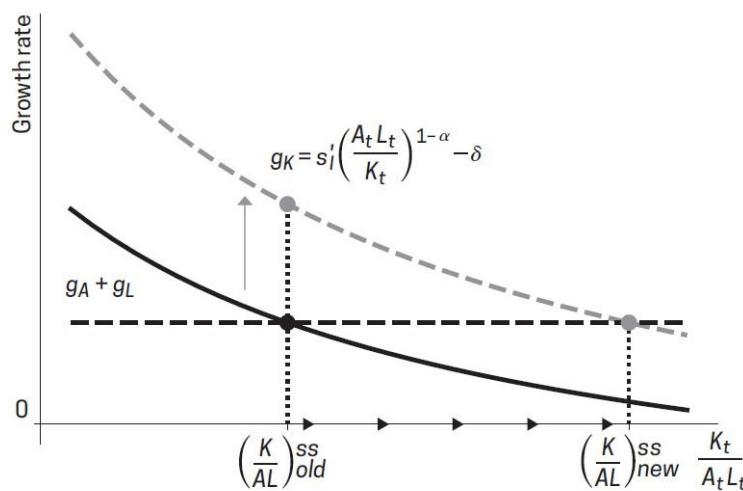
(Hint: What's the *total* effect of taxes on capital accumulation?) [15 points]

Answer. The effective saving rate is now $\hat{s} \equiv s + \tau(1 - s) > s$:

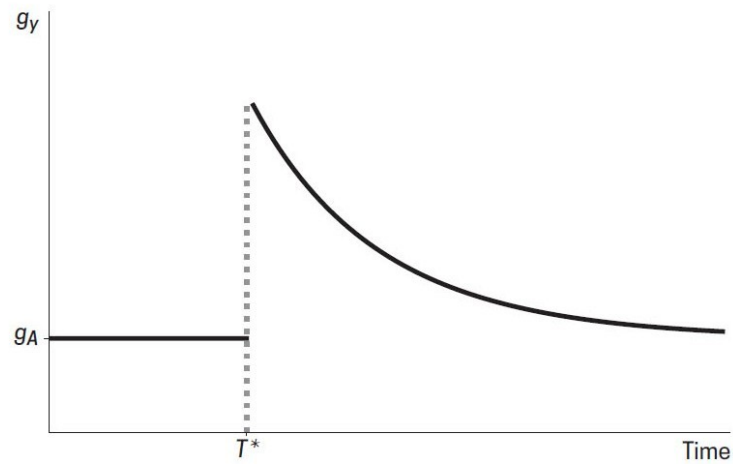
$$\begin{aligned}\dot{K}_t &= s(1 - \tau)Y_t + \tau Y_t - \delta K_t \\ &= [s - s\tau + \tau]Y_t - \delta K_t \\ &= \hat{s}Y_t - \delta K_t.\end{aligned}$$

This is the same problem that we saw in class (Week 2, Slides 29–31).

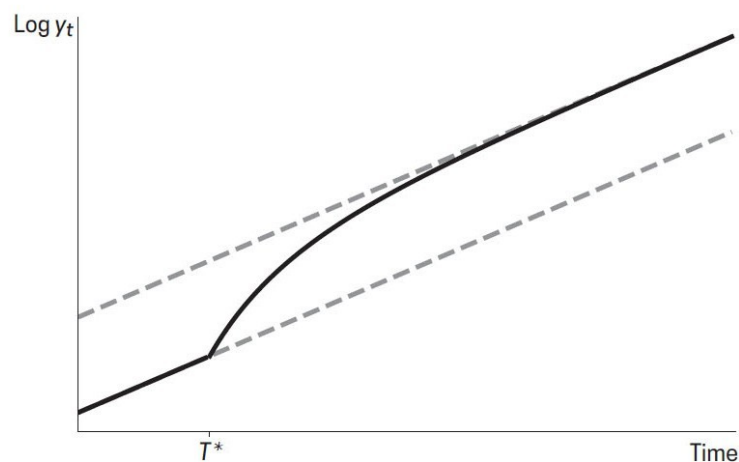
Figure 2.3 The Dynamics of an Increase in s_I



NOTE: The dark line plots the growth rate of capital, g_K , from equation (2.11) against the $K_t/A_t L_t$ ratio for the old situation. The gray line plots g_K when the capital formation rate increases to s_I' . The growth rate of productivity and labor, $g_A + g_L$, does not change. In response to the increase in the capital formation rate, the ratio $K_t/A_t L_t$ moves over time from the old steady state to the new, higher, steady state.

Figure 2.4 The Growth Rate of GDP per Capita after Change in s_I 

NOTE: In the plot, the shift to s'_I happens at time T^* . At that point the growth rate of GDP per capita jumps, but that increase is only temporary as the economy approaches the new steady state. Eventually the growth rate returns to equal g_A .

Figure 2.5 The Level of GDP per Capita after Change in s_I 

NOTE: The y-axis measures the log of GDP per capita, and the x-axis measures time. So the slope of the line indicates the growth rate of GDP per capita. The dark line plots the actual path of GDP per capita in response to the shift to s'_I at time T^* . The gray dashed lines indicate the hypothetical balanced growth paths at s_I (the lower one) and s'_I (the higher one).

3. **Why doesn't capital flow from rich to poor countries?** (Lucas, 1990) This question will walk you through another very famous paper by Bob Lucas using the neoclassical growth model. Throughout, we'll assume that $g_L = 0$ (no population growth), $g_A = 0$ (no productivity growth), and $\delta = 0$ (no depreciation) for every country i . [40 points]

- (a) Start by assuming that the representative firm in each country uses a Cobb-Douglas production function with Hicks-neutral productivity:

$$Y_i = F(K_i, L_i) = A_i K_i^\alpha L_i^{1-\alpha}.$$

Rewrite this production function in *per worker* terms (use lowercase letters for per worker variables). [5 points]

Answer. $y_i = A_i k_i^\alpha$.

- (b) Use the firm's first-order condition to derive a relationship between capital per worker k_i and the interest rate r_i . [5 points]

Answer. $r_i = \alpha A_i k_i^{\alpha-1} = \alpha \frac{y_i}{k_i}$.

- (c) Consider two countries: the US and India. Suppose we know $y_{IN}/y_{US} = 1/10$. Use your answers to (a) and (b) and a value of $\alpha = 1/3$ to determine the ratio of interest rates r_{IN}/r_{US} that this model would predict if $A_{IN} = A_{US}$. [5 points]

Answer. From (a), $k_i = (y_i/A_i)^{1/\alpha}$. Then from (b),

$$r_i = \alpha A_i k_i^{\alpha-1} = \alpha A_i \left(\frac{y_i}{A_i} \right)^{\frac{\alpha-1}{\alpha}} = \alpha A_i^{\frac{1}{\alpha}} y_i^{\frac{\alpha-1}{\alpha}}$$

$$\frac{r_{IN}}{r_{US}} = \frac{y_{IN}}{y_{US}}^{\frac{\alpha-1}{\alpha}} = \left(\frac{1}{10} \right)^{\frac{1/3-1}{1/3}} = 10^2 = 100.$$

- (d) Based *strictly* on your answer to (c), if you were an international investor choosing in which country to install your capital, would you put it in India or the US? From your casual observation of the world, does this seem to be what most investors choose to do in reality? [5 points]

Answer. Rate of return is 100× higher in India, so I'd put it in India! But in reality, capital doesn't flow from rich to poor countries: Indian investors buy US stocks, but US investors aren't routinely buying stakes in Indian companies.

- (e) Suppose we went to the Penn World Tables and found that the US actually has 8 times as much capital per worker as India: $k_{US}/k_{IN} = 8$. What would a naive development accountant conclude about the ratio of Hicks-neutral productivities A_{US}/A_{IN} ? Incorporating this conclusion, how does your prediction about the ratio of interest rates in (c) change? [5 points]

Answer. A naive development accountant would directly use the production function above to say $A_i = y_i/k_i^\alpha$, so

$$\frac{A_{US}}{A_{IN}} = \frac{y_{US}}{y_{IN}} \left(\frac{k_{IN}}{k_{US}} \right)^\alpha = 10 \times \left(\frac{1}{8} \right)^{1/3} = 10 \times \frac{1}{2} = 5.$$

Incorporating this conclusion,

$$\frac{r_{IN}}{r_{US}} = \left(\frac{A_{IN}}{A_{US}} \right)^{\frac{1}{\alpha}} \frac{y_{IN}}{y_{US}}^{\frac{\alpha-1}{\alpha}} = \left(\frac{1}{5} \right)^{1/3} \left(\frac{1}{10} \right)^{\frac{1/3-1}{1/3}} = \left(\frac{1}{5} \right)^3 \times 10^2 = \frac{100}{125} = \frac{4}{5}.$$

- (f) A wiser development accountant points out that we have not adjusted for differences in human capital between the US and India. She suggests that we instead use the production function

$$Y_i = F(K_i, h_i L_i) = A_i K_i^\alpha (h_i L_i)^{1-\alpha}.$$

Using the [Barro and Lee \(2013\)](#) education data and an estimate of the Mincer returns, she finds that the typical US worker also has 8 times as much human capital as the typical Indian worker: $h_{\text{US}}/h_{\text{IN}} = 8$. What would this wiser development accountant conclude about the ratio of Hicks-neutral productivities $A_{\text{US}}/A_{\text{IN}}$? [5 points]

Answer. The wiser development accountant would use the production function with human capital to say

$$A_i = \frac{y_i}{k_i^\alpha h_i^{1-\alpha}},$$

so

$$\frac{A_{\text{US}}}{A_{\text{IN}}} = \frac{y_{\text{US}}}{y_{\text{IN}}} \left(\frac{k_{\text{IN}}}{k_{\text{US}}} \right)^\alpha \left(\frac{h_{\text{IN}}}{h_{\text{US}}} \right)^{1-\alpha} = 10 \times \left(\frac{1}{8} \right)^{\frac{1}{3}} \left(\frac{1}{8} \right)^{\frac{2}{3}} = 10 \times \frac{1}{8} = 1.25.$$

- (g) Using your answer to (f), what would the wiser development accountant predict about the ratio of interest rates in (c)? [5 points]

Answer. From the firm's first-order condition with the new production function:

$$r_i = \alpha A_i k_i^{\alpha-1} h_i^{1-\alpha}.$$

Solving the production function for k_i and substituting in,

$$\begin{aligned} r_i &= \alpha A_i \left(\frac{y_i}{A_i h_i^{1-\alpha}} \right)^{\frac{\alpha-1}{\alpha}} h_i^{1-\alpha} \\ &= \alpha A_i^{\frac{1}{\alpha}} y_i^{\frac{\alpha-1}{\alpha}} h_i^{\frac{(\alpha-1)^2}{\alpha}} h_i^{\frac{\alpha(1-\alpha)}{\alpha}} \\ &= \alpha A_i^{\frac{1}{\alpha}} y_i^{\frac{\alpha-1}{\alpha}} h_i^{\frac{1-\alpha}{\alpha}}. \end{aligned}$$

So,

$$\begin{aligned} \frac{r_{\text{IN}}}{r_{\text{US}}} &= \left(\frac{A_{\text{IN}}}{A_{\text{US}}} \right)^{\frac{1}{\alpha}} \left(\frac{y_{\text{IN}}}{y_{\text{US}}} \right)^{\frac{\alpha-1}{\alpha}} \left(\frac{h_{\text{IN}}}{h_{\text{US}}} \right)^{\frac{1-\alpha}{\alpha}} \\ &= \left(\frac{1}{1.25} \right)^{\frac{1}{1/3}} \left(\frac{1}{10} \right)^{\frac{1/3-1}{1/3}} \left(\frac{1}{8} \right)^{\frac{1-1/3}{1/3}} \\ &= \left(\frac{4}{5} \right)^3 \times 10^2 \times \left(\frac{1}{8} \right)^2 = \frac{100}{125} \times \frac{64}{64} = \frac{4}{5}. \end{aligned}$$

This is the same as the naive prediction in (e). This actually makes sense when you recall that, whether or not the production function includes human capital, the interest rate can be written $r_i = \alpha \frac{y_i}{k_i}$, so

$$\frac{r_{\text{IN}}}{r_{\text{US}}} = \frac{y_{\text{IN}}/y_{\text{US}}}{k_{\text{IN}}/k_{\text{US}}} = \frac{1/10}{1/8} = \frac{4}{5}.$$

- (h) Based *strictly* on your answer to (g), if you were an international investor choosing in which country to install your capital, would you put it in India or the US? From your casual observation of the world, does this seem to be what most investors choose to do in reality? [5 points]

Answer. Rate of return is higher in the US, so I'd put it in the US! This is consistent with reality.

Barro, Robert J. and Jong Wha Lee. 2013. "A new data set of educational attainment in the world, 1950–2010." *Journal of Development Economics* 104:184–198.

Lucas, Robert E. 1990. "Why doesn't capital flow from rich to poor countries?" *AEA Papers and Proceedings* 80 (2):92–96.